# Sagarika Basak

Postdoctoral Research Associate

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## Education

May 2024 — PhD in Physics, Rice University, TX, USA
*Doctoral dissertation*: Quantum Phases and Dynamics of Strongly Interacting Spinor Gases in One-Dimensional Optical Lattices • *Defense:* Apr 02, 2024
• *Committee:* Dr. Han Pu (Chair), Dr. Kaden Hazzard, and Dr. Hanyu Zhu

Dec 2020 — MS (non-thesis) in Physics, Rice University, TX, USA

May 2017 — BS–MS in Physics, Indian Institute of Science Education and Research (IISER) Pune, India
• *summa cum laude* • Dean's List • CGPA: 9.1/10.0 • *Supervisor*: Dr. Rejish Nath
*Master's thesis*: Quantum Dynamics in Ultracold Rydberg Atomic Gases • *Defense:* May 02, 2017

## Research Experience

Postdoctoral Research
*Postdoctoral Research Associate*, University of Oklahoma, Norman, USA | *Advisor:* Dr. Doerte Blume

Apr 2025 – present — **Driven 4-level system** | *Co-mentoring graduate student* Luka Vilshuk
• Extending previous analysis of driven systems to 4 and higher level atomic systems
• Goal of the project is to faithfully represent the evolution of open driven N-level ladder systems
• Utilize our examinations to provide insight into the stability of symmetry protected topological systems.

May 2025 – present — **Driven 3-level system coupled to a cavity** | *Co-mentoring undergraduate student* Xylo Molenda
• Utilizing invariants to faithfully represent driven 3-level systems
• Going beyond the weak atom-cavity coupling to obtain master equations that prevail for arbitrary strength of driving and atom-cavity coupling strength.

Nov 2024 – present — **Anyons-Bosons Mixture** | *Collaboration with:* Dr. Thomas Busch
• Phase separation is seen in anyons-bosons mixtures above a certain critical interaction strength
• Preliminary analysis suggests a statistical phase sensitivity of this phase
• Exploring the utility of this sensitivity for quantum metrology

Oct 2024 – present — **Developing composite anyons** | *Collaboration with:* Dr. Thomas Busch
• Defined new "quasi"-particles: spinless anyons from spin 1/2 fermions and their governing Hubbard-type Hamiltonian
• Analyzing the ground state properties of these composite anyons and comparing it to the standard two-component anyon Hubbard model
• Preliminary results show stronger signatures of anyonic statistics
• Extending the analysis to Bose–Fermi mixtures and discussing the validity of the composite nature that presents the strong anyonic behavior

June 2024 – present — **Application of Lewis-Riesenfeld Invariants**
• Utilized the invariants to examine shortcuts to adiabaticity and equilibration in open two-level system and open harmonic oscillator
• For the driven two-level system: obtained the master equation utilizing the invariant and compared it to the master equations obtained within the laboratory, rotating and generalized frameworks
• Demonstrated invariant based master equation goes beyond the secular approximations constraining the driving field and provides an alternative to the generalized framework
• Extending the analysis to a driven open three-level system

May 2024 – July 2025 **Characterizing spin and charge transport in cold atomic setups** | *Collaborators:* Dr. Han Pu *&* Dr. Xi-Wen Guan
• Analyzing Fermi-Hubbard model quenched with a linear spin or density dependent gradient potentials that is experimentally realizable using cold atomic setups
• Extracting and examining spin and charge Drude weights
• Demonstrating spin–charge coupling with spin imbalance induced pseudo-magnetic field

May 2024 – present **Benchmarking GESC for anyons** | *Collaborator:* Dr. Han Pu
• Successful benchmarking of the ground state properties of multi-component anyons
• Using the developed formalism recreating the dipole oscillations, impurity expansion and impurity released in a linear force field observed previously using the Hubbard model (*see below*)

Doctoral Research
*Welch Fellow*, Rice University, Houston, TX, USA | *Advisor:* Dr. Han Pu

Jun 2023 – May 2024 **Ground state and quenched dynamics of multi-component anyons**
• For different interaction strengths and statistical parameters
• Extension of the generalized effective spin-chain formalism (*see below*)

Jan 2023 – May 2024 **Generalized formalism to study Bose– and Fermi–Hubbard models: Application**
• Studying large spin ground state of fermions *&* bosons, quench dynamics of spin-½ Fermi gas
• DMRG and TDVP algorithms were developed and optimized for modelling the system
• Currently examining the ground state extended SU(N) Fermi gas

Nov 2021 – Dec 2022 **Generalized effective spin-chain formalism for spinor gases: Development**
• Developed a formalism to study strongly interacting spinor gases in 1D optical lattices
• Spinor gas mapped to a system of spinless fermions and a spin-chain and developing a generalized effective spin-chain Hamiltonian that acts on the mapped system
• Significant ease of implementation with (1) generalizability: to systems of arbitrary spin statistics, (2) simple mathematical formalism, (3) universality: applicable to both lattice and continuum systems, (4) validity for arbitrary spin-dependent interaction and (5) computational efficiency

Jun 2021 – Oct 2021 **2D two-component Bose–Hubbard model with inter-component coupling with complex tunneling and coupling parameters**
• Extension to higher dimension, complex tunneling parameters to study spin phases
• Observed row-dependent spin ordering and better control of phase-types and transition behavior

Jan 2020 – May 2021 **Numerical study of the spin phases of two-component Bose–Hubbard Model**
• Mean-field treatment, exact diagonalization for small system sizes, and variational Monte Carlo (VMC) with stochastic minimization on entangled plaquette states (EPS)
• Obtained spin phases reveal novel spin behavior in the presence of coupling

Sep 2020 – Feb 2021 **Developed and optimized a variational Monte Carlo (VMC) algorithm**
• Developed a numerical technique for the study of the effective Hamiltonian beyond the mean-field approximation and for large system sizes
• System description using EPS that captured correlations better than the mean-field treatment
• To study EPS, a VMC algorithm developed *&* optimized followed by stochastic minimization

Jun 2019 – Dec 2019 **Effective spin Hamiltonian of two-component Bose-Hubbard model with inter-component coupling in the deep Mott regime**
• Effective low energy Hamiltonian was developed by mapping the Bose gas to a spin system and performing a canonical transformation followed by Schrieffer–Wolff transformation
• The Hamiltonian revealed novel correlations arising due the coupling

Jan 2018 – May 2019 **Study of strongly interacting two-component Bose-Hubbard model with inter-component coupling using spin-wave expansion**
• Literature review of the two-component Bose gas in the absence of coupling
• Developed procedure of spin wave expansion developed for spin systems
• Obtained mean field ground state in the presence of coupling, and the fluctuations above the mean field state via spin wave expansion
• Demonstrated potential of dramatic changes to spin phase space in the presence of coupling

Computational Physics Project

*Project Student*, Rice University, Houston, TX, USA | *Advisor:* Dr. Frank Toffoletto

Feb 2018 – Apr 2018 **Simulation of electromagnetically induced transparency (EIT) and dark state polaritons**

- Literature review of EIT and Dark state polaritons
- Numerical replication of susceptibility (linear and non-linear) and pulse propagation in Fortran
- Extended to include Rydberg atoms to study interacting photon– combination of EIT and Rydberg interactions, and four-level scheme system

Graduate Research Project

*Rice Fellow*, Rice University, Houston, TX, USA | *Advisor:* Dr. Kaden Hazzard

Sep 2017 – Dec 2017 **Truncated Wigner approximation (TWA) for spin systems**

- Studied Truncated Wigner Approximation, its formalism for spin systems and recent applications
- Replicated spin behavior using TWA and designed possible application for Rydberg systems

Undergraduate (5 year BS–MS)

*Master's Thesis Student*, IISER Pune, MH, India | *Advisor:* Dr. Rejish Nath

Feb 2017 – May 2017 **Periodically driven array of Rydberg atoms**

- Studied an array of single Rydberg atoms driven by a temporally modulated atom-field detuning
- Periodic modulation effectively modifies the Rabi coupling, revealing unprecedented dynamics in the presence of Rydberg-Rydberg interactions
- Observed blockade enhancement, antiblockades, and state-dependent population trapping
- Dynamical localization of a many-body configuration in a driven Rydberg lattice studied

Nov 2016 – Feb 2017 **Excitation dynamics for dissipative Rydberg atoms**

- Excitation dynamics of a system described by two interacting two-level atoms studied
- Novel envelope behavior observed, explored for dissipation and for different initial states
- Extended to three-level atoms, time-dependent detuning and time-dependent Rabi frequency

Oct 2016 – Dec 2016 **Three-body Interactions**

- 3-level Rydberg dressed interactions previously only considered up-to two-body interactions
- The significance and the profile of the three-body interactions were studied for three-level atoms in various configurations, and compared with two-body interactions
- No extraordinary significance or applications for effective interaction for such systems emerged

Sep 2016 – Oct 2016 **Rydberg dressed interactions for four-level atoms**

- 3-level lambda system extended to 4-level coupling metastable ground state to Rydberg state
- Effective interactions between two such four-level atoms investigated
- Results presented at the Okinawa School in Physics: Coherent Quantum Dynamics at OIST, Japan

Aug 2016 **Angular dependence of van der Waals coefficient**

- Previously, all the studies were done for radially symmetric van der Waals coefficient
- I considered angular dependence of van der Waals coefficient, and effective interaction computed

Jan 2016 – Aug 2016 **Rydberg dressed interactions for three-level atoms**

- Engineering interactions in ultra-cold atoms opens various possibilities to explore exotic quantum many-body phases
- Analyzed in detail a proposal for generating Rydberg-dressed interactions, based on a scheme with electromagnetically induced transparency, utilizing metastable states of alkaline earth atoms

*DAAD WISE Fellow*, Humboldt-Universitat, Berlin, Germany | *Advisor:* Dr. Alejandro Saenz

May 2015 – Aug 2015 **Ultracold Rydberg atoms in optical lattices**

- Examined potential trapping of Rydberg atom simultaneously in ponderomotive and optical trap
- Performed numerical simulations with systematic parameter variations and fidelity estimations

*Semester Project Student*, IISER Pune, MH, India | *Advisor:* Dr. Rejish Nath

Aug 2014 – Nov 2015 **Rydberg atoms in curved space-time**

- Energy states of an atom change (not unanimously) in the presence of curved space-time
- Examined the change in dipole interaction potential of Rydberg atoms in curved space-time

*Summer Project Student*, Indian Institute of Technology (IIT), Delhi, India | *Advisor:* Dr. S. K. Saha

May 2014 – Aug 2014 **Understanding the dynamics of a system of particles**
• Classical mechanics understanding of coordinate systems, motion of a particle and a rigid body
*Summer Project Student*, National Thermal Power Corporation Limited (NTPC), Noida

May 2013 – Jul 2013 **X-Ray diffraction (XRD) of coal samples** | *Advisor:* Dr. V. Jayan
• Compositional analysis of coal samples from thermal power plants using X-Ray diffraction (XRD)
• Acquiring skills in XRD techniques and research in industrial settings

May 2013 – Jul 2013 **Application of nanotechnology in coal plants** | *Advisor:* Dr. S. S. Parihar
• Nanotechnology literature review, focused on potential applications in thermal power plants
• Identified pipe scale build-up problem as targettable by nanotechnological strategies
• Assessed & designed solutions: coating internal surfaces, nano-polymers water-additives

## Grants, Honors, & Awards

Jun 2025 $650 GFB Travel Award (APS) for DAMOP 2025

Mar 2024 2023 Outstanding Reviewer for Journal of Physics Communications

Jun 2023 Graduate Student Travel Award, Department of Physics and Astronomy, Rice University
$300 travel award to attend DAMOP 2023

Jun 2023 Women in Natural Sciences Travel Award, Rice University
$500 travel award to attend DAMOP 2023

Jun 2023 $600 DAMOP Student Participation Grant

Mar 2023 2022 Outstanding Reviewer for Journal of Physics: Condensed Matter

Mar 2023 $1,000 APS Braslau Family Travel Grant for the APS March meeting.

Jun 2021 $130 DAMOP Student Participation Grant

Jun 2019 $500 DAMOP Student Travel Grant

May 2018 Robert A. Welch Fellowship
Funding from the Welch Foundation to support Research Assistantship.

Sep 2017 Sam and Helen Worden Fellowship
$2,000 awarded to a small number of incoming first-year graduate students in recognition of their past academic performance and potential for outstanding graduate work at Rice University

May 2017 Dean's List (BS-MS with Distinction)
Awarded to the top (15/109) graduates of the Dual Degree BS-MS Program at IISER Pune

Aug 2015 DAAD (*Deutscher Akademischer Austausch Dienst*, Germany) Young Ambassador
Honorary appointment as an ambassador for German university and exchange programs

May 2015 DAAD WISE (Working Internships in Science and Engineering) Fellowship
€1,825 ($2,190) for research visit to Dr. Alejandro Saenz's group at Humboldt-Universität zu Berlin

Aug 2012 INSPIRE Scholarship for Higher Education, Dept. of Science and Technology, Govt. of India
₹400,000 ($8,200) to pursue higher education in the basic sciences

Jun 2012 Top 0.37% All India Engineering Entrance Examination (AIEEE)
Central Board of Secondary Education, Ministry of HRD, Govt. of India

May 2012 Top 1.18% (All India) Joint Entrance Examination (JEE)
Indian Institute of Technology (IIT), Ministry of HRD, Govt. of India

## Research Communication

Updated list: https://scholar.google.com/citations?user=yNYx1KcAAAAJ

### Peer-Reviewed Publications

2024 Luo, J., **Basak, S.**, Pu, H., & Guan, X. W.(2025). Quantum transport in the one-dimensional Hubbard model: Drude weights and Seebeck effect. Physical Review B, **112**, 235414. doi:10.1103/wl94-p2cd

2023 **Basak, S.**, & Pu, H. (2023). Generalized effective spin-chain formalism for strongly interacting spinor gases in optical lattices. Physical Review A, **104**(5), 053326. doi:10.1103/PhysRevA.108.063315

2021 **Basak, S.**, & Pu, H. (2021). Strongly interacting two-component coupled Bose gas in optical lattices. Physical Review A, **104**(5), 053326. doi:10.1103/PhysRevA.104.053326

2018 **Basak, S.**, Chougale, Y., & Nath, R. (2018). Periodically driven array of single Rydberg atoms. Physical Review Letters, **120**(12), 123204. doi:10.1103/PhysRevLett.120.123204

Presentations

2026 **Basak, S.**, Javadi, A., & Blume, D. (2026). Lewis–Riesenfeld invariant-based master equation for driven open quantum systems. 57th Annual Meeting of the APS DAMOP.

2026 Vilshuk, L., **Basak, S.**, & Blume, D. (2026). An Algebraic Approach to Solving $n$ -Level Open Quantum System Problems with Dynamical Invariants. 57th Annual Meeting of the APS DAMOP.

2026 Molenda, X., **Basak, S.**, & Blume, D. (2026). Invariant-Based Master Equation Approach for a Driven Atom-Cavity System. 57th Annual Meeting of the APS DAMOP.

2025 **Basak, S.**, Hidalgo-Sacoto, R., Blume, D., & Pu, H. (2025). Exploring Multi-Component Anyons in 1D Optical Lattices: Statistical Phase Sensitivity and Novel Many-Body Behaviors. 56th Annual Meeting of the APS DAMOP.

2025 **Basak, S.**, & Blume, D. (2025). Exploring the driven open multi-level system using Lewis-Riesenfeld invariants. 56th Annual Meeting of the APS DAMOP.

2025 Pu, H., Luo, J., **Basak, S.**, & Guan, X. (2025). Universal charge and spin Drude weights in one-dimensional Fermi-Hubbard model. 56th Annual Meeting of the APS DAMOP.

2025 **Basak, S.** (2025) *Beyond Traditional Frameworks*: Master Equations for Driven Open Quantum Systems. Keck Retreat, University of Oklahoma.

2024 **Basak, S.** (2024) Quantum Phases and Dynamics of Strongly Interacting Spinor Gases in One-Dimensional Optical Lattices. OIST-QSU Seminar, OIST, Japan.

2023 **Basak, S.** (2023) A Generalized Effective Spin-Chain Formalism. Rice Quantum Group Meeting Seminar, Rice University.

2023 **Basak, S.**, & Pu, H. (2023). Generalized effective spin chain Hamiltonian of strongly interacting spinor gases in optical lattice. 54th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP).

2023 **Basak, S.**, & Pu, H. (2023). Study of strongly interacting spinor gases in an optical lattice using a generalized effective spin chain Hamiltonian. APS March Meeting 2023.

2021 **Basak, S.**, & Pu, H. (2021). Strongly Interacting Two-component Coupled Bose Gas in 1D Optical Lattice. 7th Smalley-Curl Institute Summer Research Colloquium.

2021 **Basak, S.**, & Pu, H. (2021). Strongly Interacting Two-component Coupled Bose Gas in 1D Optical Lattice. 52nd Annual Meeting of the APS DAMOP.

2019 **Basak, S.**, & Pu, H. (2019). Spin Phases of Strongly Interacting Two-Component Bose–Hubbard Model. 50th Annual Meeting of the APS DAMOP.

2018 Chougale, Y., **Basak, S.**, Talukadar, J., Nath, R., & Ramos, T. (2018). Coherent quantum dynamics of Rydberg atoms in different lattice geometries. 2018 C3QS: Coherent Control of Complex Quantum Systems, OIST, Japan.

2018 Chougale, Y., **Basak, S.**, Talukadar, J., Nath, R., & Ramos, T. (2018). Coherent quantum dynamics of Rydberg atoms in different lattice geometries. 2018 QMAT: National Conference on Quantum Condensed Matter, IISER Mohali, India.

2016 **Basak, S.**, Niranjan, A., & Nath, R. (2016) "Atomic Interactions in 4-level Rydberg EIT scheme based on numerical studies on effective interactions of 4-level atoms at steady state. Okinawa School in Physics: Coherent Quantum Dynamics, OIST Japan.

2016 **Basak, S.**, & Nath, R. (2016). Interaction between Resonant Rydberg dressed atoms. Workshop on Cold Atoms' Physics, IISER Pune.

### Theses

2024 **Basak, S.** (2024) Quantum Phases and Dynamics of Strongly Interacting Spinor Gases in One-Dimensional Optical Lattices (Doctoral dissertation), Rice University, May 04, 2024. Available at Rice Research Repository.

2017 **Basak, S.** (2017) Quantum Dynamics in Ultracold Rydberg Atomic Gases (Master's thesis), IISER Pune, March 20, 2017. Available at IISER Pune Digital Repository.

### *In Preparation*

2026 **Basak, S.**, Javadi, A., *&* Blume, D. (2026). Invariant-based master equation applied to driven qutrit coupled to a bath and a leaky cavity. *Under review at Physical Review B.*

2026 **Basak, S.**, *&* Pu, H. (2026). Multi-component anyons in optical lattice. *In preparation.*

2026 **Basak, S.**, *&* Blume, D. (2026). Generalized master equation using Lewis-Riesenfeld invariants of a three-level system. *In preparation.*

## Teaching

Aug 2025 – Dec 2025 *Instructor of Record*, Homer L. Dodge Department of Physics *&* Astronomy (P*&*A), University of Oklahoma, Norman, OK, USA
• PHYS 4803 Quantum Mechanics II, Fall 2025

Feb 2026 *Guest Lecturer*, P*&*A, University of Oklahoma, Norman, OK, USA
• PHYS 4183 Electricity *&* Magnetism II, Spring 2026

Jan 2018 – Dec 2019 *Teaching Assistant*, Department of Physics and Astronomy, Rice University, Houston, TX, USA
Responsible for course office hours and grading of assignments and examinations for:
• PHYS 201 Waves and Optics, Fall 2019, *Instructors:* Dr. Barry Dunning and Dr. Emilia Morosan
• PHYS 416/517 Computational Physics, Spring 2019, *Instructor:* Dr. Frank Toffoletto
• PHYS 201 Waves and Optics, Fall 2018, *Instructors:* Dr. Barry Dunning and Dr. Emilia Morosan
• PHYS 202 Modern Physics, Spring 2018, *Instructor:* Dr. Randall Hulet

## Service to the profession

Reviewer: (*Journals*) Physical Review Research, Journal of Physics Communications, Journal of Physics: Condensed Matter, Quantum Science and Technology, Physica Scripta.

Conference Session Chair:

June 5, 2026 *Low-Dimensional Bose Gases*, 57th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP)

June 18, 2025 *Light-Matter Interaction I*, 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP)

Mentorship:

Apr 2025 – present Co-mentoring, with Dr. Blume, graduate student Luka Vilshuk with their graduate research activities.

May 2025 – June 2026 Co-mentoring, with Dr. Blume, undergraduate student Xylo Molenda with their research activities as part of a Research Experiences for Undergraduates.

Oct 2024 – July 2025 Co-mentored and collaborated with visiting graduate student Raúl Hidalgo-Sacoto from Dr. Busch's group at OIST, Okinawa

June 2024 – Nov 2024 Co-mentored, with Dr. Blume, undergraduate student Madeline G. van Staveren with their research activities as part of a Research Experiences for Undergraduates.

Workshop Co-Organizer:

May 27–28, 2026 2026 Teacher Quantum Days at OU: Teaching Excellence for the Next Generation Quantum Information Science *&* Technology Workforce, Department of Physics and Astronomy, University of Oklahoma, Norman, OK, USA. *Team:* S. Basak, G. Biedermann, D. Blume, J. Gordon, and J. Stoppel.

• Co-organized a two-day quantum education and workforce-development workshop for Oklahoma high-school STEM teachers, with classroom-ready activities, hands-on demonstrations, lab tours, and quantum workforce programming.

May 16–17, 2016 — Workshop on Cold Atoms' Physics, Department of Physics, IISER Pune, MH, India
• Helped as student volunteer in organizing this two-day workshop on recent trends in cold atoms' physics led by Dr. Rejish Nath, Dr. Umakant Rapol, and Dr. Sunil Mukhi.

Professional Affiliations (*non-selective*): American Physical Society (APS), 2017–present.

## Professional Training

Jul 7 – 22, 2025 — Qiskit Global Summer School
IBM Quantum Events, *Virtual*.

Oct 14 – Nov 20, 2024 — Research visit to foster collaboration with Dr. Thomas Busch's Quantum Systems Unit
Okinawa Institute of Science and Technology, Okinawa, Japan.

Sep 29–30, 2021 — Quantum Creators Prize Symposium
University of Chicago (*Virtual*).

May 21–23, 2019 — Workshop on New Frontiers in Cold Molecules
Institute for Theoretical Atomic, Molecular and Optical Physics (ITAMP), Harvard University.

May 16–17, 2016 — Workshop on Cold Atoms' Physics @ Indian Institute of Science Education & Research, Pune.
This was on theoretical aspects of cold atoms' especially Rydberg atoms. Presented my study of effective interactions for two three-level atoms using Schmidt decomposition.

## References

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